Business Problem

Project Context

This project is based on warehouse inventory management using 7 interconnected tables:

1. Inventory Discrepancy Log
2. Inventory Item Master
3. Inventory KPI Summary
4. Inventory Profiling Dataset
5. Inventory Validation Log
6. Inventory Movement Log
7. Task Allocation Log

The main objective of this project is to analyze warehouse inventory operations and identify major business problems affecting inventory accuracy, warehouse efficiency, task completion, and staff workload management. In logistics and supply chain management, inventory accuracy is extremely important because even small errors can create major operational and financial losses.

This requires table to be merged and analyzed using Python libraries like Pandas, Seaborn, and Matplotlib.

Multiple univariate, bivariate, and multivariate analyses were conducted to derive business insights and recommendations.

The following major business statements were analyzed:

1. Which warehouses are creating the most discrepancies and why?
2. Are incomplete tasks causing discrepancies?
3. Are high-movement items more likely to create discrepancies?
4. Are some warehouses handling too much work?
5. Are some staff handling too much work?

Business Statement – 1

Which Warehouses Are Creating the Most Discrepancies and Why?

Analysis Performed -

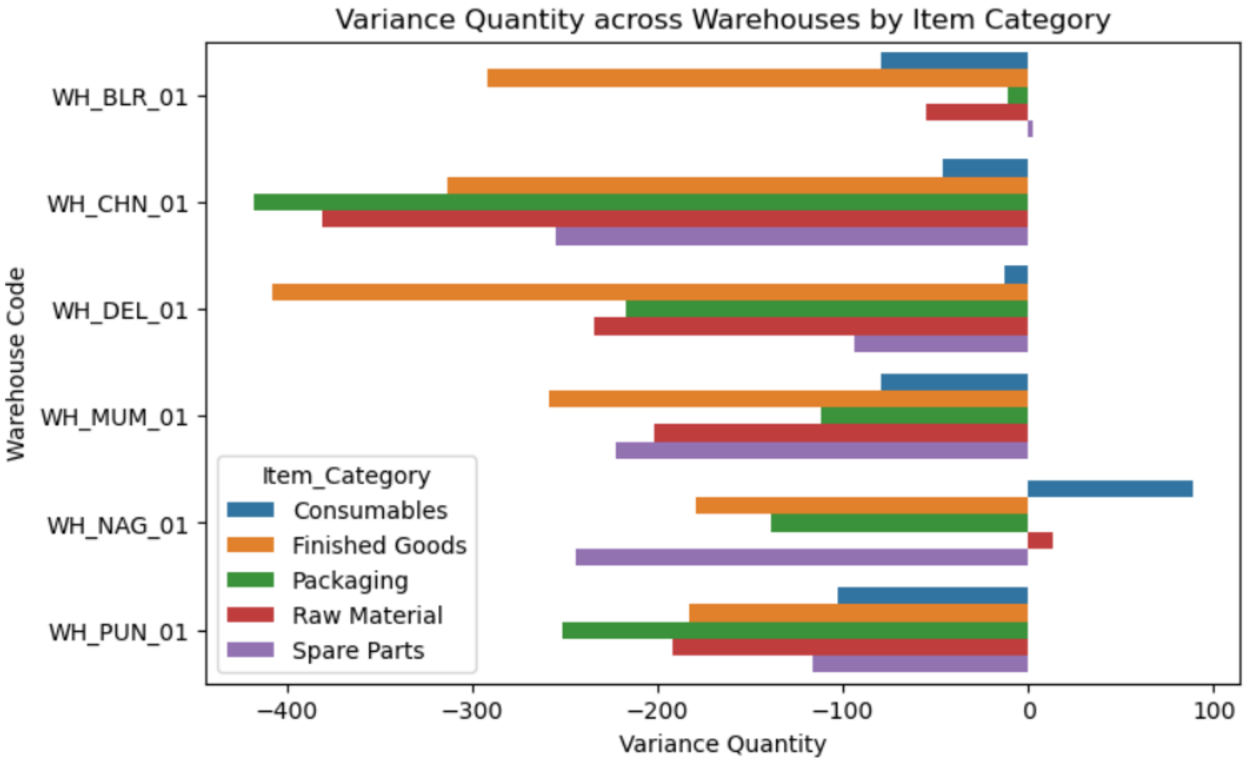
The first step was to identify warehouses with the highest discrepancy levels by grouping warehouse-wise variance quantity from the discrepancy log.

It was found that:

- WH_CHN_01 had the highest negative variance (-1413)
- WH_DEL_01 had the second highest discrepancy (-966)
- WH_MUM_01 also showed major discrepancy (-873)
- WH_PUN_01 and WH_NAG_01 followed after that

This indicates that **Chennai** warehouse is facing the highest inventory mismatch problem.

Further analysis was done by merging discrepancy data with item master data to identify which item categories create more discrepancies.



Major findings:

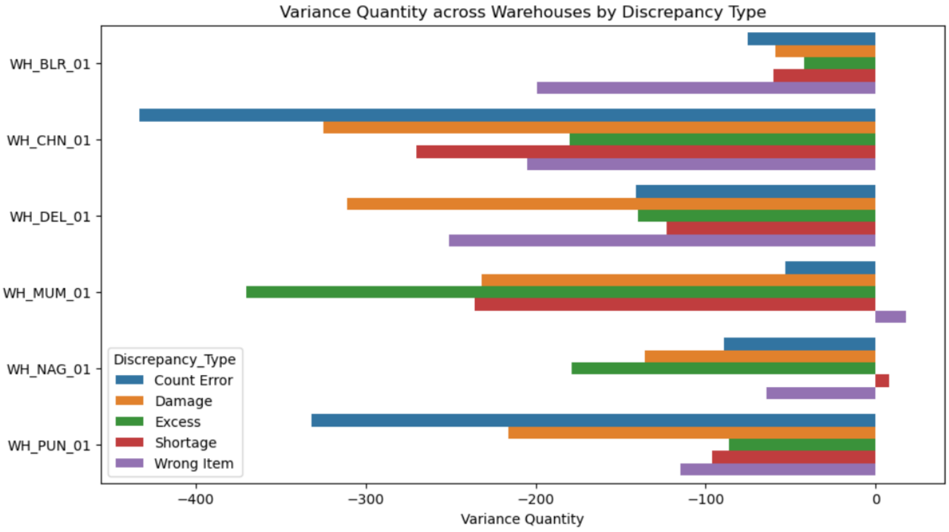
• Finished Goods showed high discrepancies in almost all warehouses

• Packaging and Raw Material categories also showed major negative variance

• WH_CHN_01 had especially high discrepancy in Packaging and Raw Material

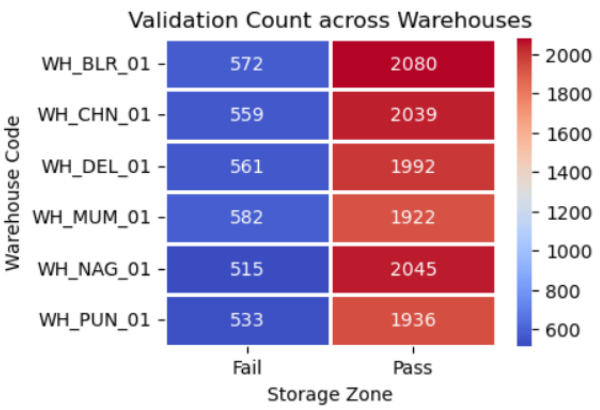
Discrepancy type analysis showed –

- Wrong Item, Count Error, Shortage were among the most frequent discrepancy types.



Validation analysis showed:

- Higher validation failures were seen in warehouses with higher discrepancies
- WH_MUM_01 and WH_BLR_01 showed significant validation failures



Warehouse_Code	Total_Movements	Variance_Quantity
WH_BLR_01	53928420	-20880
WH_CHN_01	51409224	-67824
WH_DEL_01	51042129	-46368
WH_MUM_01	50210208	-41904
WH_NAG_01	51543040	-22080
WH_PUN_01	48634362	-40560

	Warehouse_Code	Total_Movements	Variance_Quantity	Variance_%
0	WH_BLR_01	53928420	-20880	-0.038718
1	WH_CHN_01	51409224	-67824	-0.131930
2	WH_DEL_01	51042129	-46368	-0.090843
3	WH_MUM_01	50210208	-41904	-0.083457
4	WH_NAG_01	51543040	-22080	-0.042838
5	WH_PUN_01	48634362	-40560	-0.083398

KPI comparison showed discrepancy percentage against total movements:

WH_CHN_01 had the highest variance percentage (-0.13%)

Business Insight -

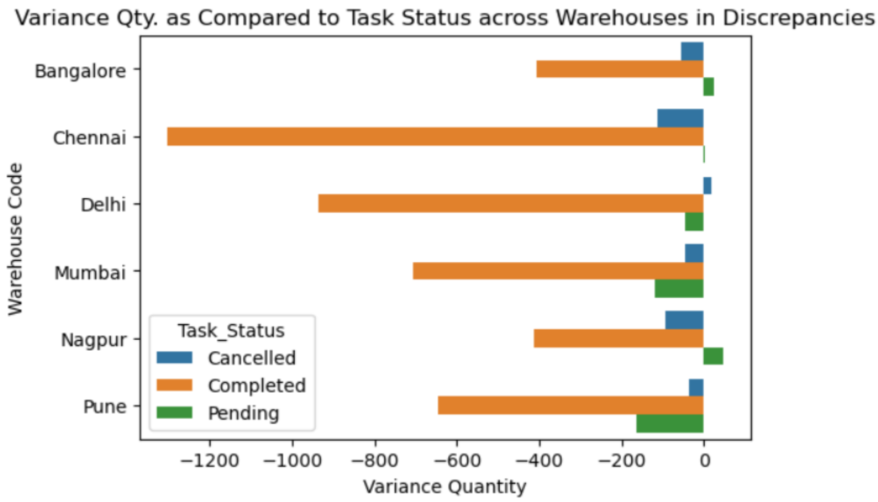
Warehouses with poor validation success and high movement pressure are creating more discrepancies. Finished Goods and Packaging require stronger monitoring.

Business Statement – 2

Are Incomplete Tasks Causing More Inventory Discrepancies?

Analysis Performed -

Discrepancy, movement, and task allocation tables were merged to study the relationship between task status and discrepancy quantity.



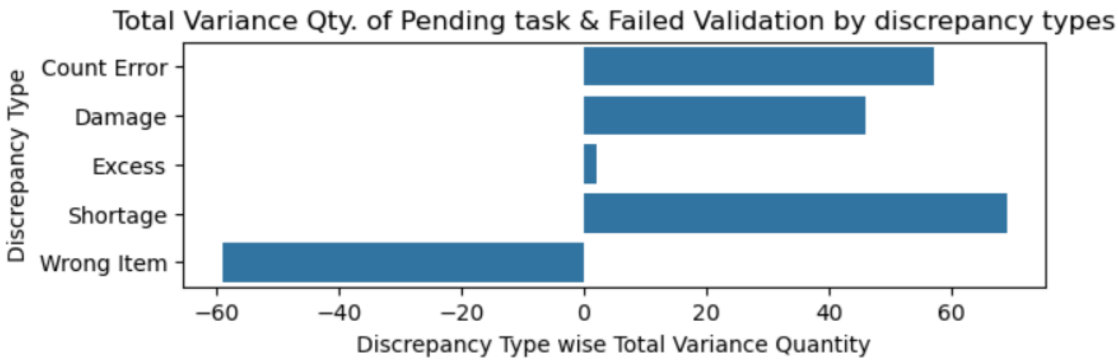
Results showed -

Completed tasks had the highest total discrepancies simply because they were highest in volume. However, pending tasks showed repeated unresolved discrepancies in multiple warehouses. Pune and Mumbai showed high discrepancy values even in Pending task status.

Next, validation status was added -

Special focus was given to: Pending Task + Validation Fail

This combination was considered dangerous.

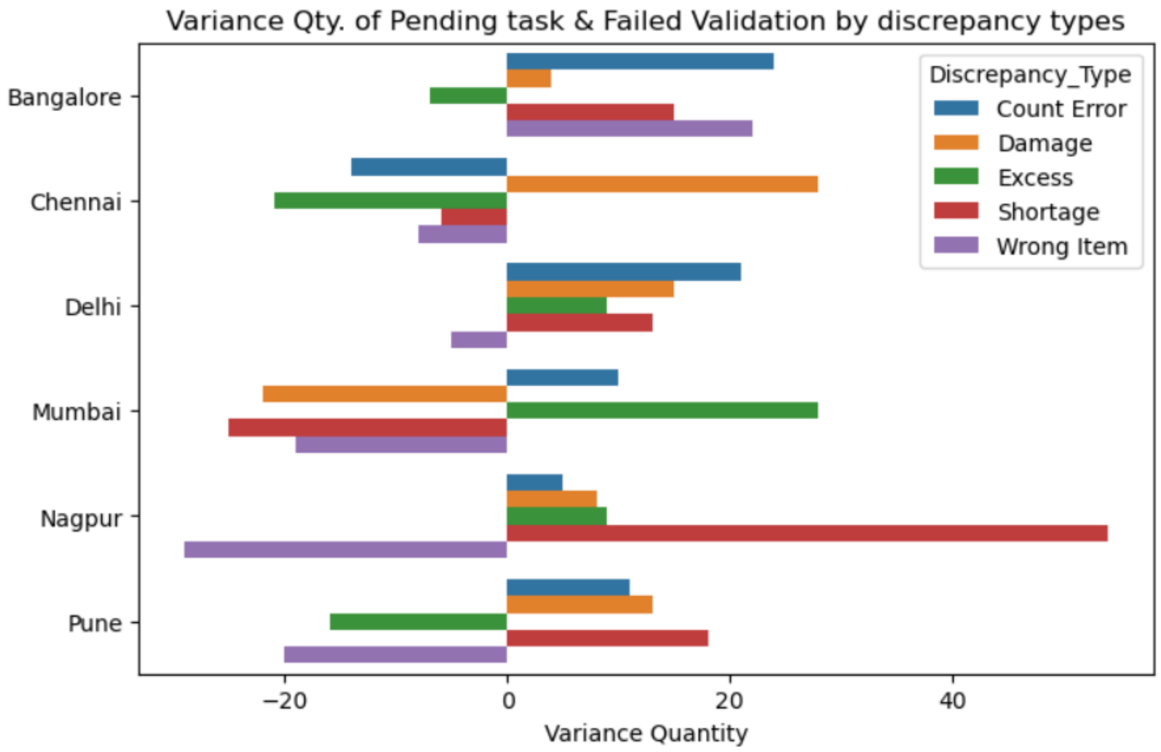


Major discrepancy types found under this condition:

- Shortage (+69)
- Count Error (+57)
- Damage (+46)

Warehouse-wise analysis showed:

- Nagpur had highest shortage problem
- Bangalore had high wrong item and count error issues
- Pune had high shortage and damage problems



Business Insight -

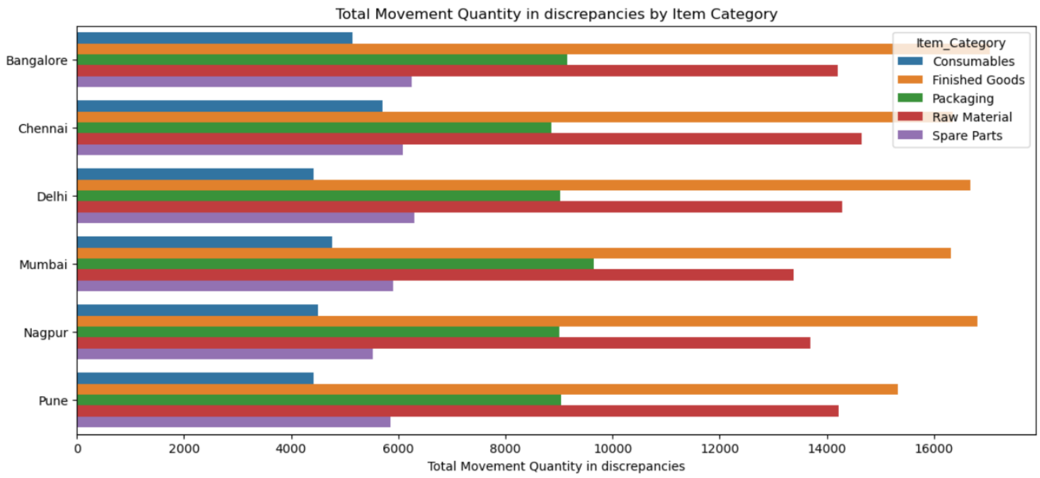
Pending tasks with failed validation create the highest operational risk. These tasks must be prioritized immediately to prevent inventory loss.

Business Statement – 3

Are High-Movement Items More Likely to Create Discrepancies? And Which items are high-risk and need special monitoring?

Analysis Performed -

Movement quantity was analyzed against item category and priority level.



Results showed:

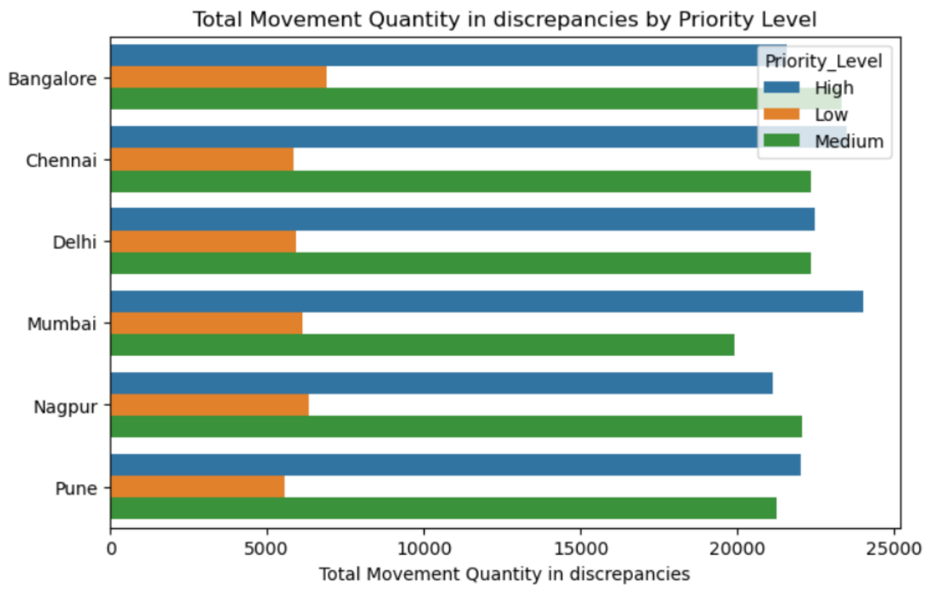
Highest movement categories were:

- Finished Goods
- Raw Material

across almost all warehouses.

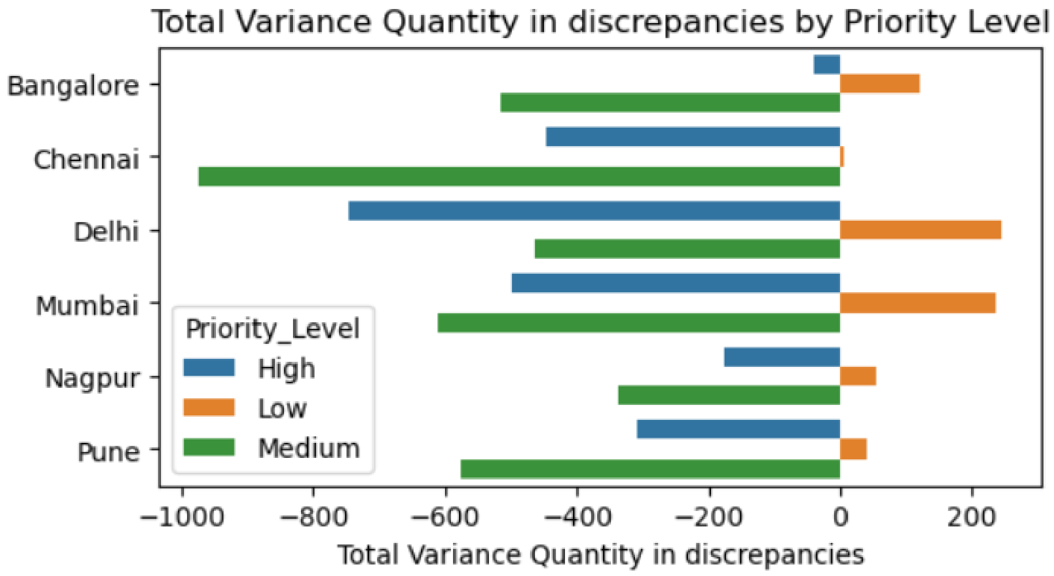
Priority level analysis showed:

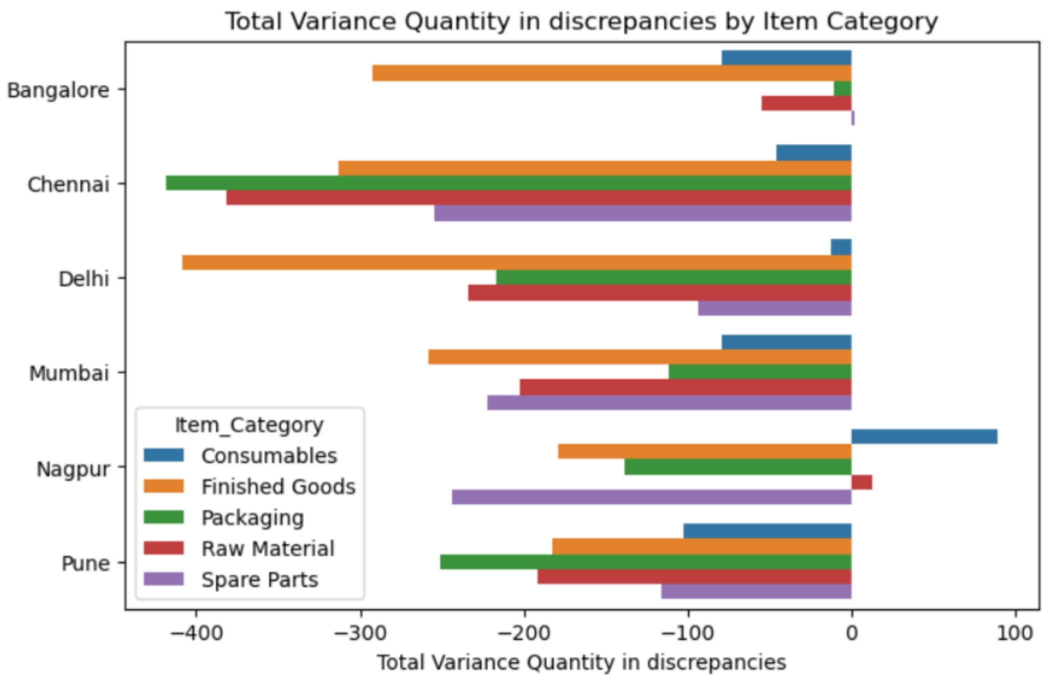
- High Priority items had the highest movement in Mumbai, Chennai, and Pune
- Medium Priority items also showed consistently high movement



Variance quantity analysis showed:

- High Priority items also had high discrepancy values
- Finished Goods showed highest variance across multiple warehouses





Business Insight -

High-movement items are more likely to generate discrepancies because of frequent handling, faster movement, and operational pressure. These items need stricter monitoring and stronger validation.

Business Statement – 4

Are Some Warehouses Handling Too Much Work?

Analysis Performed -

Movement quantity and assigned task quantity were analyzed warehouse-wise.

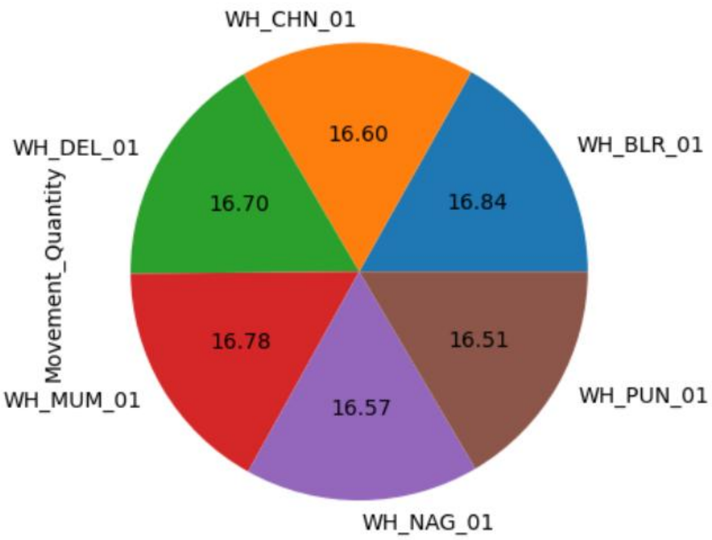


Fig (A) Warehouse Wise Movement Qty.

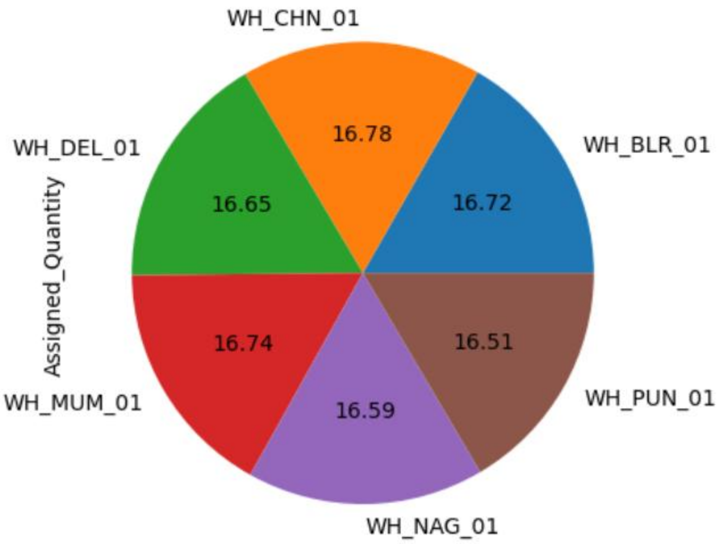


Fig (B) Warehouse Wise Assigned Qty.

Pie chart analysis showed:

- Bangalore
- Mumbai
- Delhi

These warehouses were handling slightly highest inventory movement volumes.

- Task allocation also showed similar workload concentration.
- This indicates slightly workload imbalance across warehouses.

Business Insight -

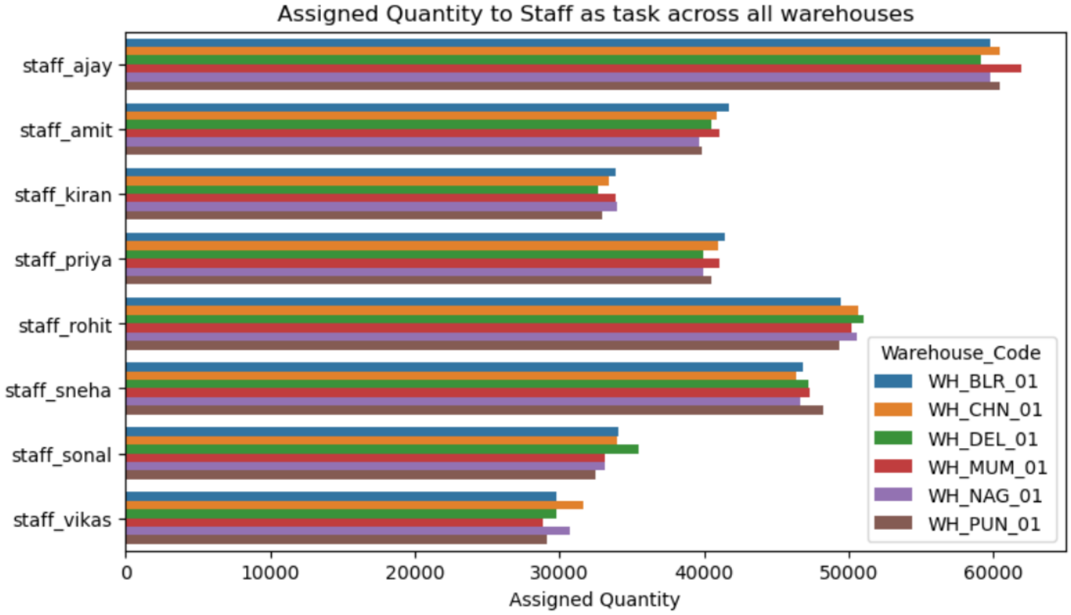
Some warehouses are slightly overloaded while others are underutilized. This creates minor stress, delays, and increases chances of discrepancies.

Business Statement – 5

Are Some Staff Handling Too Much Work?

Analysis Performed -

Staff-wise assigned quantity and movement quantity were analyzed -

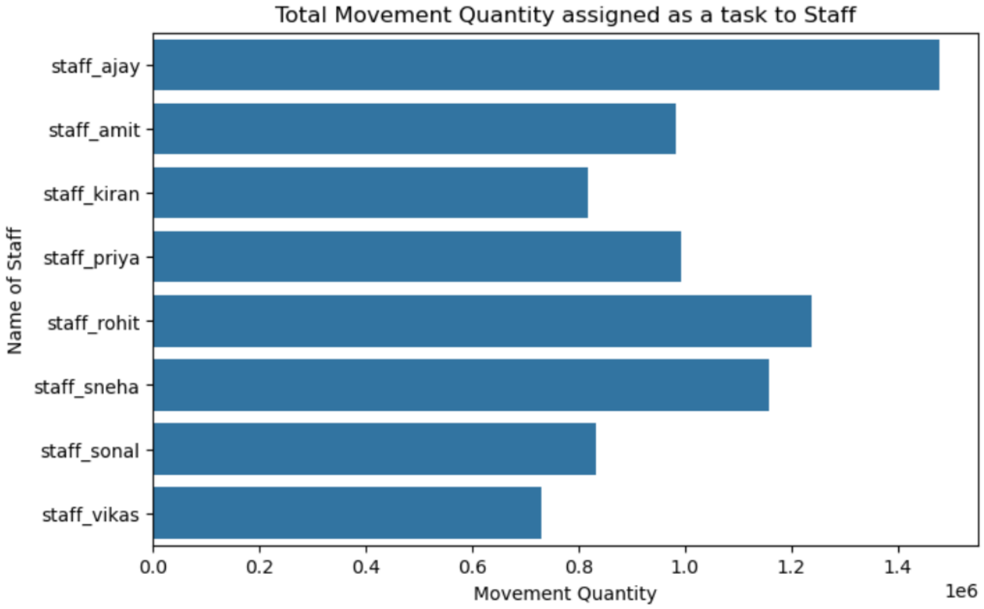


Major findings:

- staff_ajay handled the highest workload in almost every warehouse
- staff_rohit and staff_sneha also handled significantly high task loads
- staff_vikas and staff_sonal had comparatively lower workload

Movement quantity analysis also confirmed the same pattern :

This shows strong workload imbalance among staff members.



Business Insight -

Uneven staff workload creates fatigue, reduces accuracy, and increases chances of operational mistakes.

Final Recommendations

Based on the analysis:

- ✓ Improve validation checks in high-risk warehouses
- ✓ Monitor Finished Goods and High Priority items closely
- ✓ Prioritize pending tasks with failed validation
- ✓ Redistribute workload across warehouses
- ✓ Balance task allocation among staff members
- ✓ Implement stronger audit systems for discrepancy prevention

These improvements can significantly enhance warehouse efficiency and reduce operational losses.